



Python for Fixed Income

COURSE 6 · NEURAL NETWORKS

One-Dimensional Convolutions and Many Tenors

WEEK 3 · DAY 3

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Three ways to read a window

Dense, day 11

Every day its own weight: the window laid flat.

Recurrent, day 12

Day by day, a state carried forward.

Convolution, today

Five weights slid along the window.

Past tense: the 10 Yr change in bp on a date, from the 20 trading days before it.

Allowed names

fitted change in bp

walk-forward error

the last value carried forward

Never called

a forecast or a prediction

tomorrow's move or a call

an outlook or a view on rates

A filter sliding over time



20 daily changes of the 10 Yr before 2015-02-03; a filter of width 5 covers days 1 to 5, then 2 to 6, ...

```
output[t] = sum of window[t + k] * weight[k] for k = 0 to 4, plus the bias  
outputs: 20 - 5 + 1 = 16          kernel shape (width, channels, filters): (5, 1, 4)
```

Five weights of 0.2 are a 5-day moving average

First output -2.8 bp; equal to pandas' `rolling(5).mean()` within 1e-12.

A network fits its own weights: each filter is a moving sum with weights of its choosing.

A filter is a weighted moving sum

	Day 1	Day 2	Day 3	Day 4	Day 5
Change, bp	-8.0	-7.0	-1.0	7.0	-5.0
Weight	0.351	-0.4134	0.0233	0.4566	-0.4714

Same five weights at every step

Day 11's dense network gave each day its own weight; a filter has 5 however long the window.

Products added, plus a bias of 0: 5.6165

11 tenors as channels: kernel (5, 11, 4)

A filter is a block of 5 days by 11 tenors, 55 weights. Each output is one SUMPRODUCT over a rectangle; the output is still 16 steps per filter.

Keras computes a cross-correlation: the kernel is not flipped.

Pooling: one number per filter

7 of 16

filter 1's outputs set to 0 by ReLU

2.8086

average pooling, filter 1

9.5347

max pooling, filter 1

```
keras.layers.Conv1D(16, 5, activation="relu") # (20, 1) -> (16, 16)
keras.layers.GlobalAveragePooling1D()       # (16, 16) -> (16,)
keras.layers.Dense(1)                       # (16,) -> 1, a change in either sign
```

113

weights and biases, one tenor

913

weights and biases, 11 tenors

Walk-forward: the Conv1D on the 10 Yr

Model, 10 Yr alone	MAE, bp	Folds below zero change
zero change	3.21	-
training mean	3.22	-
dense, day 11	3.35	0 of 5
SimpleRNN, day 12	3.31	0 of 5
LSTM, day 12	3.23	0 of 5
GRU, day 12	3.22	1 of 5
Conv1D, today	3.24	0 of 5

Best epoch by fold, Conv1D

1, 50, 1, 1, 50

Fitted changes spread 0.25 bp against the actual 4.33 bp: little to fit, so it stays near the training mean.

Same five folds, same settings, reseeded every fold: days 11 and 12 rebuilt here.

Many tenors as channels

Network	10 Yr alone, bp	11 tenors, bp	Folds below zero change
dense	3.35	3.87	0 of 5
LSTM	3.23	3.23	1 of 5
Conv1D	3.24	3.26	0 of 5

Zero change on the same folds: 3.21 bp.

The 11-tenor Conv1D is not bit-identical run to run in its last digits (its MAE agreed to 6 decimals): printed as an MAE to 2 decimals and fold counts only.

The flattened dense network got worse; the shared-weight networks barely moved.

Same-day changes stay out

10 Yr change against	Same date	One date earlier
2 Yr change	0.75	0.02
30 Yr change	0.93	-0.03

Known only after the day

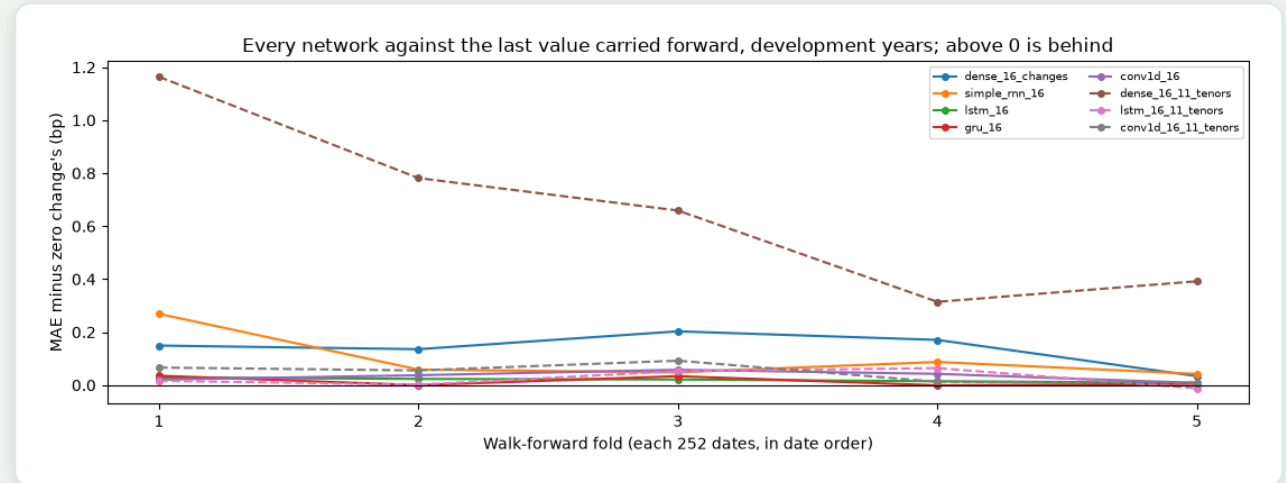
A window that ends on its target date holds other tenors' same-day changes.

```
netkit.assert_windows_past_only(changes.index[positions - 1], target_dates) # passes
netkit.assert_windows_past_only(changes.index[positions], target_dates)
# AssertionError: 1980 of 1980 windows end on or after the date they are scored on
```

make_windows: all 1,980 windows end the trading day before their target date.

Every sequence model so far

Model	Ch.	MAE, bp	Below
zero_change	1	3.21	0
training_mean	1	3.22	0
dense_16_changes	1	3.35	0
simple_rnn_16	1	3.31	0
lstm_16	1	3.23	0
gru_16	1	3.22	1
conv1d_16	1	3.24	0
dense_16_11_tenors	11	3.87	0
lstm_16_11_tenors	11	3.23	1
conv1d_16_11_tenors	11	3.26	0



Zero change is the lowest line; no network beat it on average.

The candidates for day 14

3.22 bp

gru_16, 10 Yr alone: lowest mean MAE

3.23 bp

lstm_16, 10 Yr alone: second lowest

3.21 bp

zero change, beside them on day 14

The rule

The two lowest mean walk-forward MAEs over the five development folds, on either input; every 11-tenor network was above its 10 Yr version. Chosen on development dates only; no held-out date fitted or scored.

Chosen is not good: neither beats zero change, and the seed spread is day 14's.

Excel side by side

Excel side by side (tested in Excel)

`=SUMPRODUCT(B2:B6,$K$2:$K$6)+$K$7`

`=SUMPRODUCT(B2:B6,$M$2:$M$6) = =AVERAGE(B2:B6)`

`=MAX(0,C2)`

`=AVERAGE(D2:D17) and =MAX(D2:D17)`

`=SUMPRODUCT(B2:L6,$N$2:$X$6)+$N$7`

C2 to C17: filter 1, first 5.6165

five weights of 0.2: a moving average

ReLU, filled down

pooling: 2.8086 and 9.5347

11 tenors: a 5 by 11 rectangle

Equal to netkit within 1e-12 and to Keras within 1e-5. Excel cannot fit the weights, and has no walk-forward routine.

Save the day: the sequence log

1. Morning

Files as day 12 left them

2. Code

```
conv1d_by_hand,  
assert_windows_past_only
```

3. Log, rebuilt

Days 11 and 12, no levels row

4. Today's rows

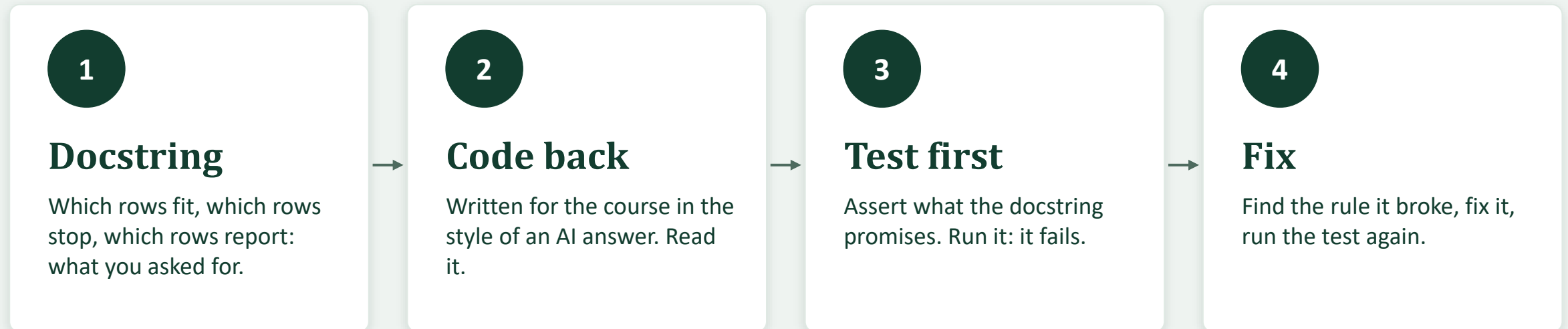
Conv1D, and 11-tenor networks

```
!git -C "{work}" log --oneline --decorate  
# (HEAD -> branch) marks the newest commit; day 14 adds a tag
```

Columns: model, channels, window, rows, folds, mae_bp, zero_change_mae_bp, folds_below_zero_change. 56 tests pass.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.



Today: 11-tenor windows end on the target date: fold 1 2.09 bp against zero change's 2.72 bp.

Code that runs is not the same as code that is right.

What you can do now

- Compute a convolution by hand; check it against Keras
- Write a filter in Excel as SUMPRODUCT filled down
- Pool a filter's outputs: average and max
- Score a Conv1D walk-forward against zero change
- Keep same-day changes out of every window

Week 3, Day 4 Case study: the held-out years, opened once